

NUNUA CHAI

nunuachai.ke

PROJECT PROPOSAL

East Africa's Creator Tipping Platform

Version: 1.0 — Draft

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Prepared By: Product & Engineering

Status: For Review

Stack: Elixir · Phoenix · Tailwind CSS · Alpine.js · LiveView (PETAL)

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1. Executive Summary

Nunua Chai is a lightweight, mobile-first creator tipping platform built specifically for Kenyan and East African creators and their audiences. The platform enables fans to send one-off monetary tips to their favourite creators — bloggers, podcasters, YouTubers, developers, artists, and educators — using M-Pesa as the primary payment rail, with zero friction and no foreign currency exposure.

Inspired by Buy Me a Coffee, Nunua Chai goes further by treating M-Pesa, KES, and the East African creator economy as first-class citizens. The 'coffee' metaphor is localised to chai, chapati, nyama choma, or any item the creator chooses. Supporters tip in Kenyan Shillings directly from their M-Pesa wallet via STK Push — no card, no bank account, and no app switching required.

The platform is built on the PETAL stack (Phoenix, Elixir, Tailwind CSS, Alpine.js, LiveView) — a battle-tested, highly concurrent, fault-tolerant foundation well-suited for real-time notifications, payment webhooks, and a lean engineering team. The initial release targets Kenya, with expansion to Tanzania, Uganda, and Rwanda planned for subsequent phases.

2. Problem Statement

2.1 The Global Platforms Don't Work Here

Platforms like Buy Me a Coffee, Ko-fi, and Patreon are built around Stripe and PayPal. Both payment processors require a Kenyan creator to have a US or EU bank account or debit card to receive funds — a barrier that effectively locks out the majority of the 30+ million M-Pesa users in Kenya.

Creators who do find workarounds face: currency conversion losses, 30-day payout delays, high international transfer fees, and a supporter experience built around USD that feels foreign to a Nairobi audience. The 'coffee' metaphor is culturally thin; a Kenyan fan sending a tip instinctively thinks in chai or nyama choma, not espresso.

2.2 The Existing Local Alternatives Are Fragmented

Many Kenyan creators resort to sharing their M-Pesa Paybill or Till numbers directly via WhatsApp or Instagram captions. This approach is untracked, unacknowledged, and offers the creator no analytics, no supporter database, and no way to say thank you programmatically. It is also invisible to fans who discover a creator later.

2.3 The Opportunity

Kenya has one of the highest mobile money penetration rates in the world. M-Pesa processed over KES 36 trillion in 2023. The Safaricom Daraja API provides a mature, well-documented integration layer. A simple, trust-worthy, creator-branded page that bridges M-Pesa to a modern tipping experience does not yet exist at scale.

3. Product Vision & Goals

3.1 Vision Statement

To be the simplest and most trusted way for East African fans to financially support the creators they love — in their own currency, on their own payment rails.

3.2 Design Principles

- Efficient — every screen has one job; no bloat, no dark patterns
- Lightweight — fast on low-bandwidth connections and mid-range Android devices
- Local-first — KES by default, M-Pesa by default, Swahili-aware UI copy
- Transparent — creators see exactly what they earn; supporters see exactly where their money goes
- Respectful — no spam to supporters, no selling of creator audience data

3.3 Success Metrics (MVP)

Metric	Target (Month 3)	Target (Month 6)
Registered creators	500	2,000
Tips processed	2,000	15,000
Total volume (KES)	KES 500K	KES 5M
Avg. tip value	KES 250	KES 300
STK Push success rate	>92%	>95%

4. Project Scope

4.1 In Scope — MVP (Phase 1)

The following capabilities constitute the Minimum Viable Product targeted for initial launch:

4.1.1 Creator Accounts

- Email + password registration; optional Google OAuth
- Unique public URL: `nunuachai.ke/@username`
- Profile: display name, avatar, short bio (280 chars), social links
- Customisable tip unit: chai, chapati, nyama choma, or a custom noun
- Customisable tip amounts: three presets (e.g. KES 50 / 100 / 250) + custom entry
- Optional thank-you message configured per creator
- Creator dashboard: earnings summary, supporter list, messages, withdrawal button

4.1.2 Supporter Tipping Flow

- No account required to tip — just a name, phone number, and M-Pesa PIN
- STK Push triggered via Safaricom Daraja API — supporter approves on their handset
- Optional message to the creator at the time of tipping
- Real-time confirmation screen upon payment success
- Creator receives in-app notification and optional SMS on each tip

4.1.3 Payments & Payouts

- All amounts in Kenyan Shillings (KES)
- M-Pesa STK Push as the sole payment method at launch
- Platform fee: 5% deducted per tip at settlement
- Creators withdraw to their personal M-Pesa via Daraja B2C API
- Minimum withdrawal: KES 100; withdrawals processed within 24 hours

4.1.4 Shareable Assets

- One-click copy of tip page URL
- Downloadable QR code (SVG + PNG) for offline use
- Embeddable HTML widget/button for external websites and blogs

4.1.5 Admin Panel

- Internal dashboard: creator list, transaction log, failed payments, withdrawal queue
- Manual override tools for disputed transactions

4.2 Out of Scope — MVP

- Recurring memberships / subscriptions (Phase 2)
- Digital product shop (Phase 2)
- Content publishing, audio posts, or newsletters (Phase 2)
- Card payments — Visa / Mastercard (Phase 2)
- Tanzania, Uganda, Rwanda expansion (Phase 2)
- Native iOS / Android application (Phase 3 — PWA first)
- Cryptocurrency payments

5. Domain Model & Bounded Contexts

The system is structured around six bounded contexts following Domain-Driven Design principles. Each context owns its data and exposes a well-defined interface to neighbouring contexts. In the Elixir/Phoenix implementation each context maps to an Elixir module namespace (e.g. `NunuaChai.Accounts`, `NunuaChai.Payments`).

Context	Responsibility	Key Entities
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Accounts	Creator registration, authentication, profile management, OAuth	Creator, Credential, OAuthToken, Session
Pages	Public creator page configuration: tip unit, presets, bio, theme, thank-you message	Page, TipPreset, Theme
Payments	Initiating STK Push, tracking payment lifecycle, webhook ingestion from Daraja	PaymentRequest, PaymentCallback, MpesaTransaction
Tips	Recording confirmed tips, linking supporter to creator, storing messages	Tip, Supporter, Message
Ledger	Creator balance tracking, fee calculation, withdrawal requests, payout via B2C	Balance, LedgerEntry, WithdrawalRequest, Payout
Notifications	Real-time and asynchronous notifications to creators and supporters via LiveView, SMS	Notification, NotificationChannel, SMSJob

5.1 Context: Accounts

Owns all identity and authentication concerns. A Creator is the primary actor — they register with an email, choose a username (used in the public URL), and configure a password. Google OAuth is supported as an alternative flow. Sessions are managed via Phoenix token-based auth with secure httpOnly cookies. This context has no dependency on any other context; other contexts reference creators by creator_id.

5.2 Context: Pages

Owns the public-facing configuration of a creator's tip page. A Creator has exactly one Page. The Page stores the localised tip unit noun (chai, chapati, etc.), up to three KES tip presets, a bio, avatar URL (stored in object storage), custom thank-you copy, and a colour theme token. LiveView renders the public page directly from Page data without additional API round-trips.

5.3 Context: Payments

The most critical context. Responsible for the full M-Pesa STK Push lifecycle: initiating the push via the Daraja OAuth + STK Push endpoint, persisting the PaymentRequest with a merchant_request_id and checkout_request_id, and receiving the asynchronous Daraja callback. An Oban background job polls for timed-out requests after 90 seconds. This context emits a payment_confirmed or payment_failed event consumed by the Tips and Ledger contexts. Webhook endpoints are secured via IP allowlisting (Safaricom IP range) and HMAC signature verification.

5.4 Context: Tips

Consumes the payment_confirmed event from the Payments context and records a persisted Tip entity linking a Supporter (identified by phone number + display name) to a Creator, along with the optional message and KES amount. The Tips context feeds the creator dashboard with paginated supporter history and powers the real-time supporter ticker on the public page via LiveView PubSub.

5.5 Context: Ledger

Maintains a double-entry ledger for each creator. On each confirmed tip, two `LedgerEntries` are written: a credit for the tip amount and a debit for the platform fee (5%). The net Balance is computed as the sum of all entries. Withdrawal requests are validated against available balance, queued as a `WithdrawalRequest`, and settled via the Daraja B2C API. The Ledger context is strictly append-only; entries are never mutated or deleted.

5.6 Context: Notifications

Handles all outbound communication. On tip confirmation the context dispatches: a `LiveView` broadcast to any open creator dashboard session (real-time supporter name and amount), and an optional SMS to the creator via an Africa's Talking or Termii gateway. Notifications are fire-and-forget from the caller's perspective; failures are logged but do not affect payment success. Jobs are processed via Oban with exponential back-off for SMS retries.

6. Technical Stack (PETAL)

The PETAL stack — Phoenix, Elixir, Tailwind CSS, Alpine.js, LiveView — is chosen for its exceptional concurrency model, fault-tolerance, and the ability to build real-time experiences without a separate JavaScript framework. For a payment-heavy product where correctness, uptime, and real-time feedback matter, Elixir/OTP is a natural fit.

Layer	Technology	Rationale
Language	Elixir 1.17	Concurrency via OTP/BEAM; fault-tolerant supervision trees ideal for payment flows
Web Framework	Phoenix 1.7	Battle-tested, HEx templates, built-in Channels + PubSub, Ecto integration
Real-time UI	Phoenix LiveView	Server-side reactive UI; real-time tip notifications without client JS framework
Styling	Tailwind CSS 3	Utility-first; tiny production CSS bundle; pairs perfectly with LiveView
Interactivity	Alpine.js	Sprinkle-on JS for modals, toasts, clipboard; no SPA overhead
Database	PostgreSQL 16	ACID guarantees for ledger entries; Ecto as ORM layer
Job Queue	Oban	Postgres-backed; reliable job processing for payment polling & SMS retries
File Storage	Tigris / S3-compatible	Avatar and QR code storage; served via CDN
Payments	Safaricom Daraja API	STK Push (C2B) for tips; B2C for creator payouts
SMS	Africa's Talking	Kenya-native SMS gateway; competitive per-SMS rates in KES

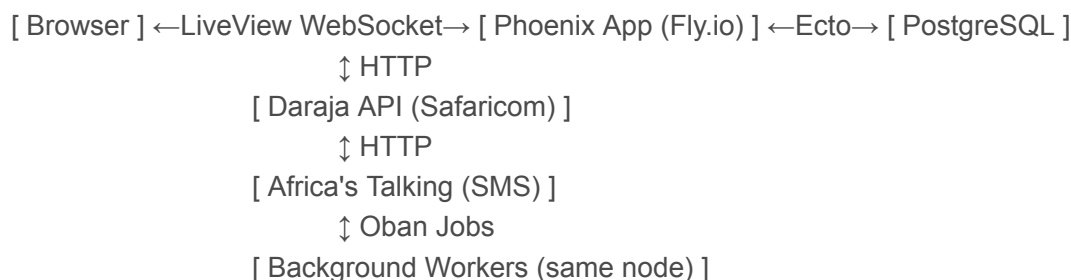
Hosting	Fly.io	Elixir-friendly; Nairobi region available; simple deploy via fly.toml
CI/CD	GitHub Actions	Mix test + Credo + Dialyzer checks on every PR; auto-deploy on main

7. System Architecture Overview

7.1 Request Flow — Supporter Tipping

- Supporter visits `nunuachai.ke/@username` — LiveView mounts the public Page
- Supporter enters name, phone, optional message, selects tip amount, clicks 'Nunua Chai'
- LiveView event → Payments context: `PaymentRequest` record created (status: pending)
- Payments context calls Daraja STK Push API; supporter receives PIN prompt on handset
- Supporter enters M-Pesa PIN — Daraja processes asynchronously (2–30 seconds)
- Daraja posts callback to `/webhooks/mpesa` — Payments context updates status (confirmed | failed)
- On confirmed: Tips context records Tip; Ledger context creates `LedgerEntries`; Notifications context fires
- LiveView PubSub broadcasts to any open creator dashboard — supporter appears in real-time

7.2 High-Level Component Diagram



7.3 Data Integrity Guarantees

- Payment confirmation and ledger entry creation are wrapped in a single `Ecto.Multi` database transaction — a tip is never recorded without a corresponding ledger credit
- Daraja callbacks are idempotent — duplicate callbacks for the same `checkout_request_id` are safely ignored via a unique index on `MpesaTransaction`
- Oban jobs run with unique constraints to prevent double-processing of payment poll jobs

8. Phased Roadmap

Phase	Timeline	Deliverables
Phase 1 — MVP	0–3 months	Creator accounts, public tip pages, M-Pesa STK Push, payouts, QR codes, embeddable widget, admin panel
Phase 2 — Growth	3–6 months	Memberships/subscriptions, digital shop, card payments (Flutterwave), Tanzania & Uganda expansion, creator analytics
Phase 3 — Scale	6–12 months	PWA / installable app, Rwanda & Ethiopia, creator newsletter tool, team accounts, white-label option

9. Risks & Mitigations

Risk	Likelihood	Mitigation
Daraja API downtime affecting tips	Medium	Oban retry with back-off; user-facing status messages; Safaricom SLA monitoring
STK Push not received by supporter	Medium	90-second timeout with clear UI feedback; manual USSD fallback instructions displayed
Fraudulent tips / chargebacks	Low	M-Pesa has no chargebacks; phone number verified at STK Push; rate limiting per phone
Safaricom API credential expiry	Low	OAuth token auto-refresh via GenServer with supervised restart; alerting on failure
Regulatory / CBK licensing requirements	Low–Medium	Legal review before launch; initially operating under Safaricom's own regulatory approvals via Daraja

10. Team & Resource Requirements

Role	FTE (Phase 1)	Responsibilities
Elixir / Backend Engineer	1.0	Phoenix app, Daraja integration, Oban jobs, Ecto schemas, API
Full-stack Engineer (PETAL)	1.0	LiveView pages, Tailwind UI, Alpine.js interactions, admin panel
Product / Design	0.5	UX flows, Figma designs, copy, creator onboarding
DevOps / Infra	0.25	Fly.io deploy, CI/CD, DB backups, monitoring

11. Indicative Budget (Phase 1)

Item	Monthly (KES)	Notes
Engineering (2 FTE × 3 months)	—	Variable by arrangement
Fly.io hosting (2 × shared-cpu-2x + Postgres)	~KES 6,500	~\$50/month
Safaricom Daraja API	Free	No monthly fee; per-transaction only
Africa's Talking SMS (est. 3,000 msgs/mo)	~KES 2,400	~KES 0.80/SMS
Object storage (Tigris)	~KES 650	~\$5/month
Domain + SSL	~KES 1,300	Annual, amortised

12. Next Steps

- Stakeholder review and sign-off on this proposal
- Apply for Safaricom Daraja API production credentials (sandbox available immediately)
- Legal review: confirm operating model does not require CBK payment service provider licence
- Finalise team resourcing and engagement contracts
- Kick off design sprint: public tip page, creator dashboard, onboarding flow
- Set up Phoenix project scaffold, Fly.io environments (staging + production), CI pipeline
- Begin Daraja STK Push integration in staging environment
- Target: internal alpha demo at end of Month 1

Nunua Chai — Simplest way to support East African creators.

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